

When Does Value-Directed Attention Matter? A Normative Model with Independent Attentional Benefit and Cost

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Abstract

An ideal observer in a spatial change-detection task has two tools for exploiting information about the reward value of cued locations: adjusting response criterion (becoming more liberal for high-value locations) and reallocating spatial attention (improving perceptual sensitivity at high-value locations at the expense of others). We develop a normative model to determine when value-directed attention (VDA)—making spatial attention track cue value—gains meaningfully over criterion adjustment alone. The model treats attentional benefit and cost as independently parameterized via an asymmetry ratio r reflecting the relative efficacy of attentional enhancement versus suppression, a distinction grounded in the dissociability of these mechanisms in primate visual cortex. Across a comprehensive parameter sweep spanning set size, cue validity, cue value, baseline sensitivity, and benefit/cost asymmetry, we find that: (1) criterion adjustment captures 60–96% of the total reward gain available from optimal allocation; (2) VDA benefit is non-monotonic in r , peaking in the cost-dominant regime where value-blind attention is too costly to deploy, creating room for value to differentiate the policy; (3) VDA is confined to a narrow parameter regime—low cue validity, high value contrast, and moderate benefit/cost asymmetry; (4) allocating less attention to the high-value location is never optimal under any parameterization. These results identify the specific conditions under which VDA and its associated neural teaching signals are behaviorally relevant, and provide a quantitative tool for designing experiments that isolate value-driven attentional effects.

1 Introduction

When sensory cues signal that some spatial locations are more rewarding than others, how should an observer allocate processing resources? This question is central to understanding both biological attention and the design of experiments that probe it. An observer informed that a change at one location is worth five times the reward of a change elsewhere could, in principle, exploit this information in two distinct ways.

The first mechanism is criterion adjustment: shifting the decision threshold at the high-value location to become more liberal, accepting more false alarms in exchange for more hits [1,2]. Criterion adjustment is “free” in the sense that it requires no perceptual reallocation—the observer simply changes how much evidence is needed to declare a change at each location. Electrophysiological evidence from primate visual cortex suggests that criterion shifts are a prominent component of how the brain responds to value cues [3,4].

The second mechanism is value-directed attention (VDA): reallocating spatial attention to the high-value location, improving perceptual sensitivity (d') there at the cost of reduced sensitivity elsewhere [5,6]. Unlike criterion adjustment, VDA involves a genuine perceptual tradeoff—resources gained at one location must be taken from another. Reward-associated stimuli can capture spatial attention [7–9], and neurons in primate visual area V4 encode reward value in a spatially specific manner that could drive such reallocation [10].

The normative question is: under what conditions does VDA gain meaningfully over optimal criterion adjustment? If criterion alone captures most of the available reward, then the neural machinery supporting value-driven attentional reallocation is redundant for this class of tasks, and experiments designed to observe VDA in this regime will produce null results. Conversely, identifying the parameter regimes where VDA is essential provides a recipe for experimental designs that can isolate value-driven attentional effects from criterion-based strategies.

A critical consideration is that attentional benefit and cost are not necessarily symmetric. Neuroscience evidence shows that attentional enhancement at attended locations—mediated by top-down feedback from frontal eye fields and intraparietal cortex—and suppression at unattended locations—mediated by lateral inhibition and surround suppression—are dissociable mechanisms that can be independently modulated [11–14]. A normative model that assumes symmetric benefit and cost may reach conclusions that are specific to that assumption rather than general properties of the task. We therefore construct a model with an explicit asymmetry ratio r that parameterizes the relative efficacy of attentional enhancement versus suppression.

We derive the conditions under which VDA is valuable and show that it is confined to a narrow regime: low cue validity, high value contrast, moderate benefit/cost asymmetry, and low baseline perceptual sensitivity. Outside this regime—which includes the operating point of most standard spatial cueing paradigms—criterion adjustment alone is sufficient.

2 Model

2.1 Task Structure

Consider a spatial change-detection task with N locations, one of which is cued. The cue conveys two pieces of information:

- Value ($v \geq 1$): the reward for detecting a change at the cued location. Changes at uncued locations are worth a reward of 1.
- Validity ($V \in [1/N, 1]$): the probability that, given a change trial, the change occurs at the cued location. Each uncued location receives probability $(1 - V)/(N - 1)$.

On each trial, a change occurs with probability 0.5 (“change trial”) or does not (“no-change trial”). The observer makes independent detection decisions at each location using signal detection theory (SDT):

$$\text{HR}(d', c) = \Phi(d'/2 - c) \quad (1)$$

$$\text{FAR}(d', c) = \Phi(-d'/2 - c) \quad (2)$$

where d' is perceptual sensitivity, c is the response

criterion, and Φ is the standard normal CDF [1, 2].

2.2 Attention Allocation

The observer allocates attention $\alpha \in [0, 1]$ to the cued location. The remaining attention is distributed equally among uncued locations, so each receives $(1 - \alpha)/(N - 1)$. At uniform attention, $\alpha = 1/N$ and all locations are treated identically.

2.3 Attention-to-Perception Mapping

Attention determines perceptual sensitivity at each location. We define a baseline transfer function $f(a) = f_0 + (1 - f_0) \cdot h(a)$, where a is the attention allocated to a location, $f_0 \in (0, 1)$ is the baseline sensitivity fraction with minimal attention, and $h(a)$ is a monotonically increasing function with $h(0) = 0$ and $h(1) = 1$. We consider four functional forms for h :

$$h(a) \in \{a, \sqrt{a}, a^{0.3}, a^2\} \quad (3)$$

spanning regimes from strongly diminishing returns ($a^{0.3}$) to accelerating returns (a^2).

At uniform attention ($\alpha = 1/N$), every location achieves a baseline sensitivity:

$$d'_{\text{base}} = d'_{\text{max}} \cdot f(1/N) \quad (4)$$

Independent benefit and cost. Attentional reallocation need not help and hurt equally. We parameterize the asymmetry between attentional benefit and cost with a ratio $r > 0$. When a location gains attention relative to the uniform baseline, its d' departure from baseline is scaled by $\beta(r)$. When a location loses attention, the departure is scaled by $\gamma(r)$:

$$\beta(r) = \frac{2r}{r + 1} \quad (\text{benefit scaling}) \quad (5)$$

$$\gamma(r) = \frac{2}{r + 1} \quad (\text{cost scaling}) \quad (6)$$

These satisfy $\beta + \gamma = 2$ (conserving total magnitude) and $\beta/\gamma = r$. The ratio r reflects the relative efficacy of attentional enhancement versus suppression. A benefit-dominant regime ($r > 1$) corresponds to neural circuits where top-down feedback is more effective at boosting attended representations than lateral inhibition is at suppressing unattended ones [12, 14]. A cost-dominant regime ($r < 1$)

corresponds to strong surround suppression at unattended locations [11]. Table 1 shows representative values.

Table 1. Asymmetry ratio r and its effect on benefit/cost scaling.

r	β	γ	Interpretation
0.1	0.18	1.82	Strongly cost-dominant
0.5	0.67	1.33	Cost-dominant
1.0	1.00	1.00	Symmetric
2.0	1.33	0.67	Benefit-dominant
10.0	1.82	0.18	Strongly benefit-dominant

When $\alpha > 1/N$ (attending to the cued location), the cued location gains attention and uncued locations lose it:

$$d'_{\text{cued}} = d'_{\text{base}} + \beta \cdot [d'_{\text{max}} f(\alpha) - d'_{\text{base}}] \quad (7)$$

$$d'_{\text{uncued}} = d'_{\text{base}} + \gamma \cdot [d'_{\text{max}} f(\frac{1-\alpha}{N-1}) - d'_{\text{base}}] \quad (8)$$

Note that $d'_{\text{max}} f(\frac{1-\alpha}{N-1}) - d'_{\text{base}} \leq 0$ when uncued locations receive less than $1/N$ attention, so the γ -scaled term reduces uncued d' . If $\alpha < 1/N$, the roles reverse: the cued location's departure is scaled by γ and each uncued location's by β . All d' values are clamped at ≥ 0 .

At $r = 1$, $\beta = \gamma = 1$, and the model reduces to a symmetric special case where a single shared transfer function governs both benefit and cost. Figure 1 illustrates the mapping for three values of r .

2.4 Reward Structure

We evaluate two reward variants to separate task-design effects from fundamental incentive structure.

Variant A (value-coupled correct rejection): The correct rejection reward $\text{CR} = V \cdot v + (1 - V)$ is coupled to value and validity, calibrated so that the expected value of responding and withholding are balanced at uniform attention. Hit at cued: v ; hit at uncued: 1; CR: $V \cdot v + (1 - V)$; false alarm or miss: 0.

Variant B (fixed correct rejection): $\text{CR} = 1$, independent of v and V . This creates a stronger marginal benefit for detection at high-value locations, since the hit-CR difference at the cued position scales with v .

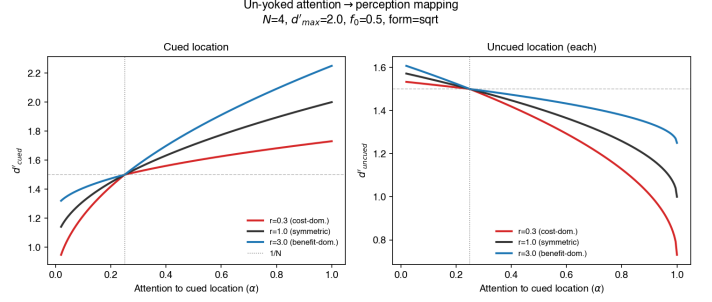


Figure 1. Attention-to- d' mapping for three benefit/cost regimes. Left: d' at the cued location as a function of α . Right: d' at each uncued location. In the cost-dominant regime ($r=0.3$, red), shifting attention incurs steep costs at uncued locations with modest cued gains. In the benefit-dominant regime ($r=3.0$, blue), attending provides large gains with minimal cost. The symmetric case ($r=1.0$, black) lies between. Dashed line: baseline d' at uniform attention. Parameters: $N=4$, $d'_{\text{max}}=2.0$, $f_0=0.5$, $\sqrt{\cdot}$ form.

Expected reward on a single trial:

$$E[R] = 0.5[V \cdot \text{HR}_c \cdot v + (1 - V) \cdot \text{HR}_u] + 0.5 \cdot P_{\text{no-fa}} \cdot \text{CR} \quad (9)$$

where $P_{\text{no-fa}} = (1 - \text{FAR}_c)(1 - \text{FAR}_u)^{N-1}$ is the probability of no false alarm on a no-change trial.

2.5 Policy Decomposition

To isolate the contribution of each mechanism, we define four nested policies:

1. Joint optimum (P1): α^* , c^*_{cued} , c^*_{uncued} jointly optimised at each v . This is the best any observer can achieve.
2. Value-blind attention (P2): α fixed at $\alpha^*(v=1)$ —the optimal allocation when all locations have equal value—then criteria re-optimised at each v . This observer uses attention for validity but not for value.
3. Uniform attention (P3): $\alpha = 1/N$, criteria optimised. This observer uses only criterion to encode value.
4. Floor (P4): $\alpha = 1/N$, $c = 0$. An unbiased baseline that uses neither mechanism.

The reward gain decomposes as:

$$\text{criterion gain} = R(P3) - R(P4) \quad (10)$$

$$\text{validity-attention gain} = R(P2) - R(P3) \quad (11)$$

$$\text{VDA benefit} = R(P1) - R(P2) \quad (12)$$

The VDA benefit is our central metric: the reward gained by making attention track cue value on top of optimal validity-driven attention and criterion. The criterion fraction— $(R(P3) - R(P4)) / (R(P1) - R(P4))$ —measures what share of total reward gain is captured by criterion alone.

3 Methods

3.1 Parameter Space

Primary sweep. We fix a reference regime ($N = 4$, $d'_{\max} = 2.0$, $f_0 = 0.5$, $\sqrt{\cdot}$ form) and sweep: r at 21 log-spaced values from 0.1 to 10.0 (plus $r = 1.0$); V at 21 values from $1/N$ to 1.0; $v \in \{1, 2, 3, 4, 5\}$; both reward variants. Total: 4,410 parameter combinations.

Secondary sweeps. At $v = 5$, $V = 0.5$: (a) $f_0 \in \{0.1, 0.3, 0.5, 0.7\}$; (b) all four functional forms; (c) $N \in \{2, 4\}$.

Crossover analysis. For each $(V, v, \text{variant})$ combination, Brent’s method finds r^* where VDA benefit crosses zero.

3.2 Optimisation

For each parameter combination, we find the $(\alpha^*, c_{\text{cued}}^*, c_{\text{uncued}}^*)$ that maximise $E[R]$ via vectorised grid search over a three-dimensional array $R[\alpha, c_{\text{cued}}, c_{\text{uncued}}]$. Grid resolution: $\Delta\alpha = 0.005$, $\Delta c = 0.05$. The α grid extends below $1/N$ (starting at 0.02) to allow the optimiser to discover inverted attention ($\alpha^* < 1/N$) if it is ever beneficial. The criterion grid auto-expands when optima lie on boundaries.

3.3 Validation

At $r = 1$, the model reduces to the symmetric special case. We verified this by checking that all 210 matched parameter combinations yield identical α^* and R^* values to machine precision (Appendix A).

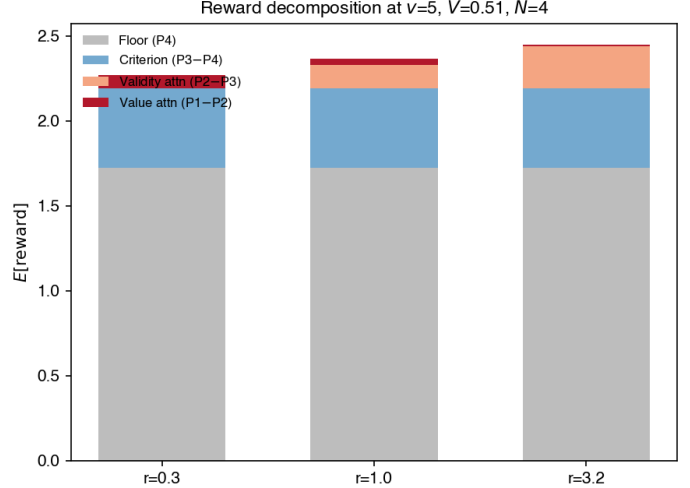


Figure 2. Reward decomposition at $v = 5$, $V \approx 0.5$, $N = 4$, variant A. Grey: floor reward. Blue: criterion gain ($R_{P3} - R_{P4}$). Orange: validity-attention gain ($R_{P2} - R_{P3}$). Red: VDA benefit ($R_{P1} - R_{P2}$). Criterion dominates across all three regimes.

4 Results

4.1 Criterion Adjustment Dominates Value Encoding

Across the full parameter space, criterion adjustment captures the majority of the reward gain available from optimal allocation. Figure 2 decomposes expected reward into its components at three representative asymmetry ratios. In the cost-dominant regime ($r = 0.3$), the criterion fraction reaches 96%—nearly all value-related reward is captured by adjusting the response threshold, with attention contributing only a thin sliver. In the symmetric case ($r = 1.0$), the criterion fraction is 73%. Even in the benefit-dominant regime ($r = 3.2$), where attention is cheap to deploy, criterion still accounts for 64% of total gain.

This pattern holds across the full sweep: the criterion fraction ranges from 60% to 96% across all (r, V, v) combinations tested. Criterion adjustment is always the single largest contributor to value-related reward.

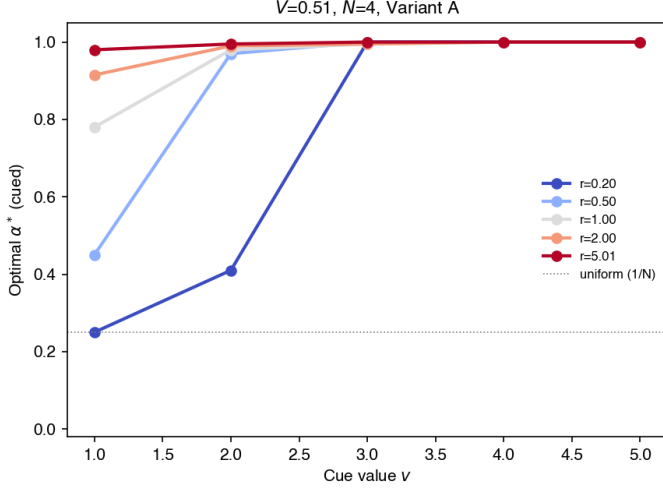


Figure 3. Optimal cued attention α^* vs cue value v at $V = 0.5$, $N = 4$, variant A. At cost-dominant r (warm colours), the value gradient drives large shifts in α^* . At benefit-dominant r (cool colours), α^* is already near ceiling for all v , leaving no room for VDA.

4.2 Benefit/Cost Asymmetry Shapes Attention Allocation

The asymmetry ratio r fundamentally determines how the value-blind observer (P2) allocates attention, which in turn determines how much room remains for value-directed attention (P1) to improve upon.

Figure 3 shows optimal α^* versus cue value v at $V = 0.5$ for several r values. At cost-dominant $r = 0.2$, the value-blind baseline P2 allocates nearly uniform attention ($\alpha \approx 1/N = 0.25$) because even the validity gradient does not justify the high perceptual cost of shifting attention. But when value is high ($v = 5$), the joint optimum P1 shifts α^* to 1.0—the $5\times$ reward makes it worth paying the cost. This P1–P2 gap is the VDA benefit.

As r increases, P2’s α rises because shifting attention becomes cheaper. By $r = 5.0$, P2 already allocates $\alpha \approx 0.98$, nearly matching P1, and the gap between them closes.

4.3 VDA Benefit is Non-Monotonic in r

Figure 4 reveals the central finding: VDA benefit is non-monotonic in r . At $V = 0.5$ and $v = 5$, VDA peaks near $r \approx 0.3$ (VDA = 0.080 reward units) and declines toward zero at both extremes.

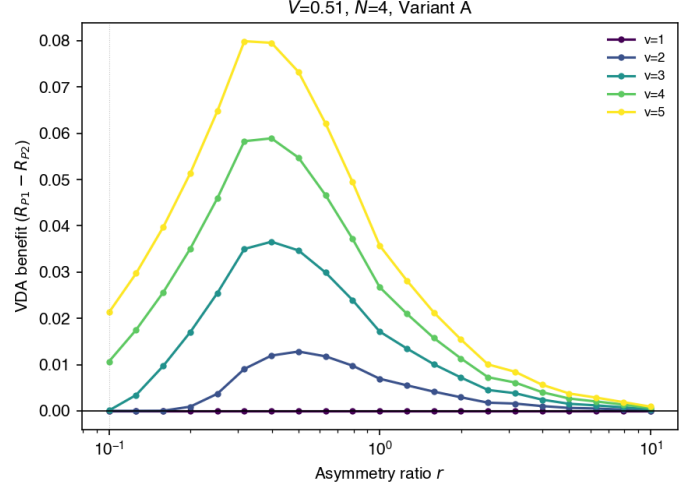


Figure 4. VDA benefit vs asymmetry ratio r at $V = 0.5$, $N = 4$, variant A. Each line represents a different cue value v . VDA peaks in the cost-dominant regime ($r \approx 0.3$) and approaches zero at both extremes.

The mechanism is a squeeze from two directions. At low r (strongly cost-dominant), shifting attention is so expensive that neither P1 nor P2 moves from uniform—both converge to $\alpha = 1/N$ and $VDA \rightarrow 0$. At high r (strongly benefit-dominant), shifting attention is so cheap that even the value-blind P2 saturates near $\alpha = 1.0$, leaving P1 no room to improve—again $VDA \rightarrow 0$. Only at intermediate r is the value-blind policy non-trivially suboptimal: P2 does not fully exploit the validity gradient (because attention still has meaningful cost), while P1 can justify the cost at high v (because the reward is large enough). This intermediate regime is where VDA has its maximal impact.

At $v = 1$ (no value gradient), VDA is exactly zero for all r , confirming that the model correctly identifies value asymmetry as the driver.

4.4 The Regime Where VDA Matters

Figure 5 maps the full strategy landscape at $v = 5$ across the (r, V) plane. Two patterns are apparent. First, at high validity ($V \geq 0.75$), optimal α^* is near 1.0 and VDA is negligible (< 0.005 reward units) regardless of r . The validity gradient alone saturates attention allocation, leaving no room for value to contribute. Second, the “hot zone” for VDA—the region where it exceeds 0.02 reward

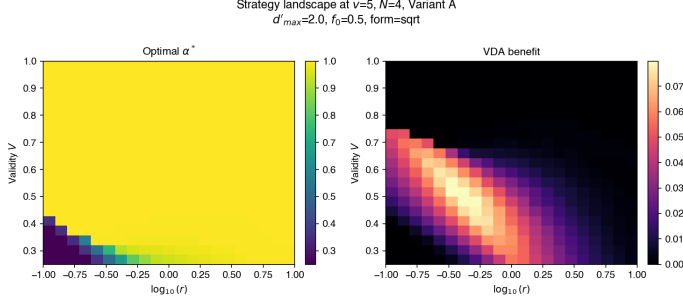


Figure 5. Strategy landscape at $v=5$, $N=4$, variant A. Left: optimal α^* . Right: VDA benefit. At high V , α^* is near 1.0 regardless of r and VDA is negligible. The high-VDA zone is concentrated at low validity and moderate r .

units—is confined to low V (near $1/N$) and moderate r (approximately 0.2–1.0).

This identifies a clear recipe for the regime where VDA matters: low cue validity (so validity alone does not saturate attention), high value contrast ($v \gg 1$), moderate or cost-dominant benefit/cost asymmetry (so the value-blind policy is sub-optimal), and, as the secondary sweeps confirm, low baseline sensitivity f_0 (so attention is perceptually consequential).

4.5 Inverted Attention is Never Optimal

Across all 4,410 rows of the primary sweep—spanning $r \in [0.1, 10]$, $V \in [0.25, 1.0]$, $v \in \{1, \dots, 5\}$, and both reward variants—the optimal α^* is always $\geq 1/N$. The α grid was extended below $1/N$ (down to 0.02), allowing the optimiser to discover inverted attention (preferentially attending away from the high-value location) if it were ever beneficial. It never is.

This result follows from the reward structure: attending away from the high-value cued location means (a) the cued location’s d' drops below baseline, scaled by the cost factor γ , while (b) uncued locations’ d' rises above baseline, scaled by the benefit factor β . Since $v \geq 1$ and $V \geq 1/N$, the weighted reward loss at the high-value cued location always exceeds the gain at low-value uncued locations, regardless of r . In the most cost-dominant regime ($r \rightarrow 0$), the optimal strategy converges to uniform attention ($\alpha^* = 1/N$), never below it.

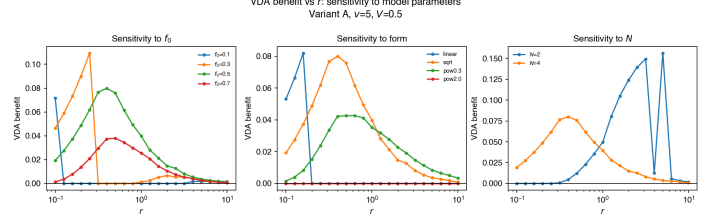


Figure 6. Sensitivity of the VDA- r relationship to model parameters at $v=5$, $V=0.5$, variant A. Left: baseline sensitivity f_0 . Centre: functional form. Right: set size N . The non-monotonic pattern is robust.

4.6 Robustness Across Model Parameters

Figure 6 shows VDA benefit versus r at $v=5$, $V=0.5$ for variations in baseline sensitivity f_0 , functional form, and set size N . The non-monotonic pattern is qualitatively robust across all variations.

Lower f_0 (attention more perceptually consequential) shifts the peak VDA higher and to slightly lower r —when attention strongly determines whether a stimulus is seen, the value of deploying it to the right location increases. The functional form has modest effects: strongly diminishing returns ($a^{0.3}$) compresses the VDA peak, while accelerating returns (a^2) stretches it. $N=2$ shows qualitatively similar patterns to $N=4$ with slightly larger VDA magnitudes, as the binary allocation decision is simpler.

5 Discussion

5.1 Why Criterion Dominates

The dominance of criterion adjustment over VDA is not an incidental feature of specific parameter choices—it reflects the fundamental asymmetry between the two mechanisms. Criterion adjustment is costless: shifting c at one location has no effect on d' at any location. It can therefore independently encode value at each location without any perceptual tradeoff. Attention reallocation, by contrast, is a zero-sum game: improving d' at the cued location necessarily degrades d' elsewhere, and this cost must be justified by a reward differential large enough to overcome it [5, 15].

The non-monotonic VDA- r relationship reveals the deeper reason: VDA only gains when the value-

blind policy is non-trivially suboptimal. At high r , attention is cheap and even value-blind allocation saturates. At low r , attention is expensive and both value-blind and value-aware policies retreat to uniform. Only in the intermediate regime does value information differentiate the policies.

5.2 Implications for Experimental Design

These results have direct practical implications for researchers designing experiments to study value-driven attentional effects.

When VDA should be observable. The model predicts that VDA contributes meaningfully to performance only when: (a) cue validity is low (near $1/N$), so that the validity gradient does not saturate attention; (b) value contrast is high ($v \gg 1$); (c) baseline sensitivity without attention is low (attention is perceptually consequential); and (d) the benefit/cost profile is moderately cost-dominant. Tasks that meet these criteria—such as low-validity spatial cueing with large reward differentials and stimuli near threshold—should produce the largest behaviorally measurable VDA effects.

When VDA should not be expected. Standard spatial cueing paradigms with high validity ($V \geq 0.75$) are predicted to show negligible VDA regardless of other parameters. In these regimes, the validity gradient alone drives attention to ceiling, and any observed value-related performance differences are more parsimoniously attributed to criterion adjustment. Experimenters should not interpret the absence of VDA in high-validity paradigms as evidence against value-driven attentional mechanisms—the normative model predicts that VDA is simply not needed in this regime.

5.3 Implications for Computational Models

For computational models of value-directed behavior, these results suggest that the architecture’s reliance on criterion-like mechanisms (response bias, threshold adjustment) versus attentional reallocation to encode value should depend on the task regime. In high-validity spatial detection tasks, a model that routes value through criterion rather than attention is normatively efficient. Models such

as recurrent vision transformers trained on spatial change-detection tasks [16] may discover this efficient routing without explicit instruction, finding that criterion adjustment captures value-related reward while attention serves primarily validity-driven functions. The attention inversion sometimes observed in such models—where attention is preferentially allocated away from high-value locations—cannot be rationalised by any benefit/cost asymmetry, as our analysis shows this is never normatively optimal.

5.4 Biological Interpretation of the Asymmetry Ratio

The parameter r can be interpreted as reflecting the relative effectiveness of two dissociable neural mechanisms: attentional gain modulation (top-down feedback enhancing representations at attended locations) and attentional suppression (lateral inhibition reducing representations at unattended locations) [11, 12, 14, 17].

Empirical measurements of attentional benefits and costs—which are often asymmetric and depend on task parameters such as stimulus-onset asynchrony, eccentricity, and set size [5, 13, 18]—could in principle locate specific experimental preparations in the model’s parameter space. This would allow quantitative predictions about when VDA should be observed in those preparations.

The finding that peak VDA occurs in the cost-dominant regime ($r < 1$) has an interesting implication: value-directed attention is most valuable precisely when attention is most costly to deploy. This suggests that if the brain maintains specialised machinery for value-driven attentional control [10, 19, 20], its utility may be greatest in perceptual conditions characterised by strong surround suppression or lateral inhibition.

5.5 Limitations

Several simplifications bound the generality of our conclusions. First, the model assumes independent per-location SDT decisions; real observers emit a single global response, introducing dependencies that could alter the optimal policy. Our results therefore represent an upper bound on VDA bene-

fit. Second, the asymmetry ratio r is a single global parameter; real neural circuits may have location-specific, feature-specific, or time-varying asymmetries. Third, the $\beta + \gamma = 2$ constraint conserves total attention magnitude; alternative constraints (e.g., multiplicative $\beta\gamma = 1$) could yield quantitatively different results, though the qualitative findings—non-monotonic VDA, no inversion, criterion dominance—should be robust. Fourth, we do not model the learning dynamics by which an observer discovers the optimal policy; the speed and reliability of learning criterion adjustments versus attentional reallocation may further favour criterion in practice [3].

6 Conclusion

Value-directed attention occupies a narrow niche in the normative landscape of spatial change detection. Across a comprehensive parameter sweep spanning set size, cue validity, cue value, baseline sensitivity, and benefit/cost asymmetry, criterion adjustment consistently captures the majority of value-related reward. VDA contributes meaningfully only when cue validity is low, value contrast is high, baseline sensitivity is poor, and the benefit/cost profile lies in a moderate, cost-dominant range. Outside this regime, criterion alone is sufficient.

Inverted attention—allocating less attention to high-value locations—is never optimal under any parameterization, establishing a clear normative boundary on rational attention allocation.

These results provide a quantitative framework for predicting when VDA should be behaviorally observable, guiding both the design of primate experiments and the interpretation of computational models of value-driven attention.

A Validation: Symmetric Special Case

At $r = 1$, the model’s independent benefit and cost scaling reduces to a symmetric special case ($\beta = \gamma = 1$) where a single shared transfer function governs both benefit and cost. We validated this by comparing the $r = 1$ results against an indepen-

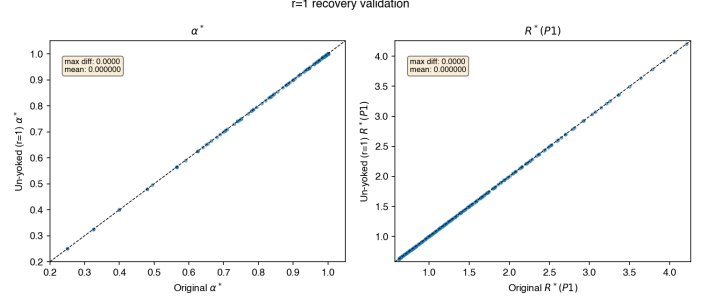


Figure 7. Validation: the model at $r = 1$ exactly recovers the symmetric special case. Left: optimal α^* . Right: optimal reward $R^*(P1)$. All 210 points fall on the identity line.

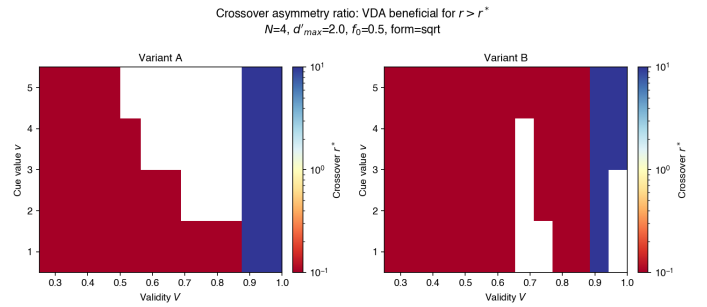


Figure 8. Crossover r^* in (V, v) space. Colour: the asymmetry ratio at which VDA transitions from zero to positive. White cells: VDA positive across the entire swept r range.

dent implementation of the symmetric model across all 210 matched parameter combinations ($N = 4$, $d'_{\max} = 2.0$, $f_0 = 0.5$, $\sqrt{\cdot}$ form). Optimal α^* and R^* values are identical to machine precision (maximum difference: 0.0; Figure 7).

B Crossover Analysis

Figure 8 maps the crossover r^* —the asymmetry ratio at which VDA benefit transitions from zero to positive—across the (V, v) parameter space. In most of the space, VDA is positive for all $r \geq 0.1$. Crossovers occur primarily at the lowest validity levels (V near $1/N$), where VDA depends critically on the benefit/cost balance, and at high validity ($V \approx 0.75$) where VDA is negligible and numerical precision limits the analysis.

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